

DIGITAL FORENSIC EXAMINATION REPORT

Mobile Device Forensics via ADB

Live Data Acquisition and Triage of a Personal Android 13 Device

ANDROID DEBUG BRIDGE (ADB) · PLATFORM-TOOLS · LIVE DEVICE TRIAGE

REPORT ID	DFCS · CCFA · ANDROID · 2026-07
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ACADEMIC PROGRAM	B.S. Digital Forensics and Cyber Security (DFCS)
CERTIFICATION TRACK	Certified Computer Forensics Analyst (CCFA)
EVIDENCE REFERENCE	Personal Android 13 smartphone (OPPO CPH2365), examiner-owned device
EXAMINATION FOCUS	Live ADB data acquisition: system info, apps, logs, network, files, and usage history
PRIMARY TOOLS	Android SDK Platform-Tools (ADB)
REPORT DATE	July 26, 2026
CLASSIFICATION	Academic Coursework: Portfolio Release Copy

This document reproduces coursework performed against the examiner's own personal Android device for digital forensics education, in a controlled lab environment. A small number of network addresses have been redacted from the source screenshots to protect the examiner's personal network identity; all other content is shown as captured.

Report Details

Field	Detail
Examiner	Awais Ahmed
Role	Digital Forensics Student / Examiner-in-Training
Source Practical	Android ADB mobile forensics worksheet
Evidence Source	Examiner's own Android 13 device (OPPO CPH2365), acquired live via ADB

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1. Executive Summary

This report documents a live mobile forensics exercise performed against the examiner's own Android 13 smartphone using the Android Debug Bridge (ADB), as part of Digital Forensics and Cyber Security (DFCS) coursework aligned to Certified Computer Forensics Analyst (CCFA) competencies. Unlike the disk-image examinations elsewhere in this portfolio, this exercise works against a live, running device connected over USB rather than a static forensic image.

The exercise covers the full ADB triage workflow: enabling and authorizing device access, pulling system identification, enumerating installed applications, capturing real-time logs, extracting battery and network state, acquiring accessible files and directories, and reviewing device usage history. It closes with a direct look at where Android 13's security model blocks further access (contacts, call logs, SMS, and protected system files), and what a rooted device or forensic-mode image would additionally make available.

A small number of screenshots in this report have been redacted. Two network addresses (a global-scope IPv6 address and a recurring public IPv4 address) were blacked out to protect the examiner's personal network identity; everything else, including account emails and the device serial number, is shown as captured, since none of it identifies anyone beyond the examiner on their own device.

2. Scope and Authorized Use

All actions in this report were performed by the examiner against their own personally-owned device, with Developer Options and USB Debugging deliberately enabled for this exercise, in a controlled lab environment. ADB access requires the device owner to physically accept an on-screen authorization prompt, meaning this kind of acquisition cannot be performed silently or without the device holder's knowledge and consent. This report should be read strictly as a record of self-directed coursework, not as guidance for accessing a device without its owner's authorization.

3. Background: ADB and Android Forensics

The Android Debug Bridge (ADB) is a command-line tool, included in the Android SDK Platform-Tools, that lets a connected workstation issue commands directly to an Android device over USB. It is a developer and debugging tool first and a forensic tool second, but because it exposes shell access, file transfer, and system service dumps, it is widely used for live, non-invasive triage of Android devices when a full physical or file-system image is not available or not necessary.

3.1 Enabling Access

ADB access is disabled by default and must be deliberately enabled on the device itself: Developer Options is unlocked by tapping the Build Number seven times under Settings → About Phone, after which USB Debugging can be turned on. The first time a new workstation connects, the device displays an authorization prompt that the owner must accept; declining it, or not responding, leaves the device listed as unauthorized rather than device when queried.

3.2 Android 13 and Scoped Access

Beginning with Android 11, and continuing through the Android 13 device examined here, Google's platform restricts ADB shell access to sensitive content providers, contacts, call logs, and SMS, and to protected system directories such as /data, unless the device is rooted or running a special forensic build. This is a deliberate privacy and security boundary, not a limitation of ADB itself, and it directly shapes which parts of this exercise succeeded and which did not, covered in Section 6.

4. Examination Environment and Tools

- Android SDK Platform-Tools (includes ADB), installed on the examiner's forensic workstation.

Android SDK Platform-Tools download: <https://developer.android.com/tools/releases/platform-tools>

- Target device: an OPPO CPH2365 smartphone running Android 13, connected via USB in File Transfer (MTP) mode with USB Debugging enabled.
- Notepad (Windows) was used to review larger exported text files (logcat output, battery statistics) once redirected to disk.

5. Procedure and Results

5.1 Device Connection and Authorization

Purpose: Confirms the workstation can see and has been granted authorized access to the target device before any data collection begins; an unauthorized device will not respond to further commands.

```
C:\Users\Awais>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
960e9609          unauthorized
```

Figure 1. Initial adb devices query, showing the device listed as unauthorized before the on-device prompt was accepted.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb devices
* daemon not running; starting now at tcp:5037
* daemon started successfully
List of devices attached
960e9609          device
```

Figure 2. adb devices confirming the device now shows as device (authorized) after accepting the USB debugging prompt.

Finding: Device serial 960e9609 initially listed as unauthorized, then device once the on-screen USB debugging prompt was accepted on the phone itself.

5.2 Basic System Information

Purpose: Establishes the device's model, OS version, and serial number, the minimum identifying information any examination report needs before presenting further findings.

adb shell getprop dumps the full Android system property list; the full dump is extensive; individual properties were then queried directly for the values that matter for reporting.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell getprop
[DEVICE_PROVISIONED]: [1]
[audio.hw_burst_min_usec]: [2000]
[audio.mmap_exclusive_policy]: [2]
[audio.mmap_policy]: [2]
[adjust.preinstall.path]: [/data/etc/appchannel/adjust.preinstall]
[af.fast_track_multiplier]: [2]
[arm64.memtag.process.system_server]: [off]
[audio.deep_buffer.media]: [true]
[audio.offload.min.duration.secs]: [30]
[audio.offload.video]: [true]
[audio.sys.mute.latency.factor]: [2]
[audio.sys.noisy.broadcast.delay]: [500]
[audio.sys.offload.pstimeout.secs]: [3]
[audio.sys.routing.latency]: [0]
```

Figure 3. adb shell getprop, showing the beginning of the full system property dump.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell getprop ro.product.model
CPH2365
```

Figure 4. Querying ro.product.model directly.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell getprop ro.build.version.release
13
```

Figure 5. Querying ro.build.version.release directly.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell getprop ro.serialno
960e9609
```

Figure 6. Querying ro.serialno directly.

Finding: Model: CPH2365 (OPPO). Android version: 13. Serial number: 960e9609.

5.3 Installed Applications

Purpose: Builds a complete inventory of installed software and where each package physically lives on the device, useful for identifying non-default or unusual applications and for correlating with other artifact categories.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell pm list packages -f
package:/data/app/---QyLIZTMf-SUSCIdEUBISw==/com.google.android.apps.walletnfcrel-aonP_dtxfe71noooe25Xmw==/base.apk=com.google.android.apps.walletnfcrel
package:/data/app/Opplote2/Opplote2.apk=com.coloros.note
package:/my_bigball/priv-app/KekMarket/KekMarket.apk=com.heyta.market
package:/system_ext/priv-app/SystemUI/SystemUI.apk=com.android.systemui
package:/system/app/BookmarkProvider/BookmarkProvider.apk=com.android.bookmarkprovider
package:/product/priv-app/OShare/OShare.apk=com.coloros.oshare
package:/system_ext/priv-app/BlockedNumberProvider/BlockedNumberProvider.apk=com.android.providers.blockednumber
package:/data/app/---YH6TzLxQydp2K4KQVwQ==/com.google.android.marvin.talkback-7KXVLUeQs3jKXlJLxPQyQ==/base.apk=com.google.android.marvin.talkback
package:/data/app/---Y9HxPvYjKIZQx2h918XQ==/com.google.android.verifier-11xW7VPAD_FH8AKwmlp-1w==/base.apk=com.google.android.verifier
package:/apex/com.android.tethering/priv-app/ServiceConnectivityResourcesGoogle@371021120/ServiceConnectivityResourcesGoogle.apk=com.google.android.connectivity.es
```

Figure 7. adb shell pm list packages -f, listing installed package paths and identifiers.

The output was then redirected to a file for easier review and preservation, rather than left only on-screen.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell pm list packages -f > installed_apps.txt
```

Figure 8. Redirecting the package list to installed_apps.txt.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>type installed_apps.txt
package:/data/app/---QyLIZTMf-SUSCIdEUBISw==/com.google.android.apps.walletnfcrel-aonP_dtxfe71noooe25Xmw==/base.apk=com.google.android.apps.walletnfcrel
package:/data/app/Opplote2/Opplote2.apk=com.coloros.note
package:/my_bigball/priv-app/KekMarket/KekMarket.apk=com.heyta.market
package:/system_ext/priv-app/SystemUI/SystemUI.apk=com.android.systemui
package:/system/app/BookmarkProvider/BookmarkProvider.apk=com.android.bookmarkprovider
package:/product/priv-app/OShare/OShare.apk=com.coloros.oshare
package:/system_ext/priv-app/BlockedNumberProvider/BlockedNumberProvider.apk=com.android.providers.blockednumber
package:/data/app/---YH6TzLxQydp2K4KQVwQ==/com.google.android.marvin.talkback-7KXVLUeQs3jKXlJLxPQyQ==/base.apk=com.google.android.marvin.talkback
package:/data/app/---Y9HxPvYjKIZQx2h918XQ==/com.google.android.verifier-11xW7VPAD_FH8AKwmlp-1w==/base.apk=com.google.android.verifier
package:/apex/com.android.tethering/priv-app/ServiceConnectivityResourcesGoogle@371021120/ServiceConnectivityResourcesGoogle.apk=com.google.android.connectivity.es
package:/data/app/---IvC4vt9ddzo0k9LUBV887w==/com.google.android.tts-XI1xI2tZHWYomoBMLKYGjw==/base.apk=com.google.android.tts
package:/data/app/---AMVUXYIuQ10FRPrWj11fsw==/idm.internet.download.manager.plus-wxwBaZOR05JyJfAnbydNYQ==/base.apk=idm.internet.download.manager.plus
package:/system_ext/priv-app/OpusLauncher/OpusLauncher.apk=com.android.launcher
package:/product/priv-app/OpusScreenRecorder/OpusScreenRecorder.apk=com.oplus.screenrecorder
```

Figure 9. Reviewing the saved installed_apps.txt with type.

Finding: Installed packages included standard OEM/system components (SystemUI, BookmarkProvider, OShare) alongside user-installed software (Google Wallet, TalkBack, IDM Internet Download Manager, OPPO Note); nothing outside the expected profile of a personal daily-use device was identified.

5.4 Real-Time and Historical Logs

Purpose: Captures the device's rolling system log buffer, which can contain crash traces, application activity, and system events not recorded anywhere else, valuable for timeline reconstruction, but only for as long as the buffer has not yet been overwritten.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb logcat -d > system_logs.txt
```

Figure 10. Redirecting adb logcat -d (dump-and-exit mode) to system_logs.txt.

The exported log was opened in Notepad rather than read on the command line, since a raw logcat dump is large and difficult to scan line by line in a terminal.

```

system_logs - Notepad
File Edit Format View Help
----- beginning of crash
06-08 03:26:59.320 620 620 F libc : Fatal signal 6 (SIGABRT), code -1 (SI_QUEUE) in tid 620 (init), pid 620 (init)
06-08 03:26:59.362 620 620 F libc : crash_dump helper failed to exec, or was killed
06-08 03:26:59.370 628 628 F libc : Fatal signal 6 (SIGABRT), code -1 (SI_QUEUE) in tid 628 (init), pid 628 (init)
06-08 03:26:59.387 628 628 F libc : crash_dump helper failed to exec, or was killed
07-11 15:14:29.362 4580 4580 F libc : Fatal signal 6 (SIGABRT), code -1 (SI_QUEUE) in tid 4580 (init), pid 4580 (init)
07-11 15:14:29.440 4580 4580 F libc : crash_dump helper failed to exec, or was killed
07-12 11:14:29.362 21245 21245 F DEBUG : Process name is /vendor/bin/hw/android.hardware.audio.service, not key_process
07-12 11:14:29.362 21245 21245 F DEBUG : keyProcess: 0
07-12 11:14:29.362 21245 21245 F DEBUG : *** *** *** *** *** *** *** *** *** ***
07-12 11:14:29.362 21245 21245 F DEBUG : Build fingerprint: 'OPPO/CPH2365/OP4F11L1:13/TP1A.220905.001/R.2377cfd-54785-54786:user/release-keys'
07-12 11:14:29.362 21245 21245 F DEBUG : Revision: '0'
07-12 11:14:29.362 21245 21245 F DEBUG : ABI: 'arm'
07-12 11:14:29.362 21245 21245 F DEBUG : Timestamp: 2026-07-12 11:14:29.000495625+0500
07-12 11:14:29.362 21245 21245 F DEBUG : Process uptime: 0s
07-12 11:14:29.362 21245 21245 F DEBUG : Cmdline: /vendor/bin/hw/android.hardware.audio.service
07-12 11:14:29.362 21245 21245 F DEBUG : pid: 1241, tid: 1241, name: audio.service >>> /vendor/bin/hw/android.hardware.audio.service <<<
07-12 11:14:29.362 21245 21245 F DEBUG : uid: 1041
07-12 11:14:29.362 21245 21245 F DEBUG : signal 35 (<debugger signal>), code -1 (SI_QUEUE from pid 1358, uid 1041), fault addr -----
07-12 11:14:29.362 21245 21245 F DEBUG : r0 00000004 r1 c0306201 r2 ffe87368 r3 ffe872bc
07-12 11:14:29.362 21245 21245 F DEBUG : r4 e7834000 r5 c0306201 r6 ffe87368 r7 00000036
07-12 11:14:29.362 21245 21245 F DEBUG : r8 00000000 r9 e7834054 r10 ffe87378 r11 ffe87368
07-12 11:14:29.362 21245 21245 F DEBUG : ip 00000100 sp ffe872a8 lr e7822083 pc e756637c
07-12 11:14:29.362 21245 21245 F DEBUG : backtrace:
07-12 11:14:29.362 21245 21245 F DEBUG : #00 pc 0009537c /apex/com.android.runtime/lib/bionic/libc.so (__ioctl+8) (BuildId: 52d57af1f3765fa26f91321f6692fa3a)
07-12 11:14:29.362 21245 21245 F DEBUG : #01 pc 0006107f /apex/com.android.runtime/lib/bionic/libc.so (__ioctl+26) (BuildId: 52d57af1f3765fa26f91321f6692fa3a)
07-12 11:14:29.362 21245 21245 F DEBUG : #02 pc 0005f789 /apex/com.android.vndk.v30/lib/libhidbase.so (android::hardware::IPCThreadState::getAndExecuteCommand
07-12 11:14:29.362 21245 21245 F DEBUG : #03 pc 0006088d /apex/com.android.vndk.v30/lib/libhidbase.so (android::hardware::IPCThreadState::joinThreadPool(bool)
07-12 11:14:29.362 21245 21245 F DEBUG : #04 pc 00002605 /vendor/bin/hw/android.hardware.audio.service (main+2564) (BuildId: b5ba2845ffd2ad1c80c85239344b8c71)
07-12 11:14:29.362 21245 21245 F DEBUG : #05 pc 00053a3f /apex/com.android.runtime/lib/bionic/libc.so (__libc_init+54) (BuildId: 52d57af1f3765fa26f91321f6692fa3a)
07-12 11:14:29.815 1358 2213 F libc : Fatal signal 6 (SIGABRT), code -1 (SI_QUEUE) in tid 2213 (TimerThread), pid 1358 (audioserver)
07-12 11:14:30.835 21277 21277 F DEBUG : Process name is /system/bin/audioserver, not key_process
07-12 11:14:30.835 21277 21277 F DEBUG : keyProcess: 0
07-12 11:14:30.835 21277 21277 F DEBUG : *** *** *** *** *** *** *** *** *** ***
07-12 11:14:30.835 21277 21277 F DEBUG : Build fingerprint: 'OPPO/CPH2365/OP4F11L1:13/TP1A.220905.001/R.2377cfd-54785-54786:user/release-keys'
07-12 11:14:30.835 21277 21277 F DEBUG : Revision: '0'

```

Figure 11. system_logs.txt opened in Notepad, showing native crash traces (SIGABRT) tied to specific process IDs and build fingerprints.

Finding: The captured buffer included several native crash events (Fatal signal 6 / SIGABRT) tied to system services such as android.hardware.audio.service, each timestamped and attributed to a specific process ID, alongside the exact build fingerprint (OPPO/CPH2365/OP4F11L1:13/...) of the device at capture time.

5.5 Battery and Power Data

Purpose: Records the device's power state at the time of acquisition, which can help establish whether a device was actively charging, idle, or under load around the time of an event of interest.

```

C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell dumpsys battery
Current OPLUS Battery Service state:
  Charger voltage : 4916
  Battery current : -330
  ChargerTechnology: 0
  ChargeFastCharger: false
  PlugType: 2
  UpdatesStopped: false
  UsbHwStatus: 0
  BatteryHwStatus: 1
  HwStatusIsSet: 1
  BatteryIcStatus: 1
  IcStatusIsSet: 1

  mUsbStatus: 0
  PhoneTemp: 330
  ThermalFeatureOn: true
Current Battery Service state:
  AC powered: false
  USB powered: true
  Wireless powered: false
  Max charging current: 500000
  Max charging voltage: 5000000
  Charge counter: 2750000
  status: 2
  health: 2
  present: true
  level: 55
  scale: 100
  voltage: 3953
  temperature: 332
  technology: Li-ion

```

Figure 12. adb shell dumpsys battery output: charger voltage, battery current, and charging state.

Finding: The device was on AC charge at the time of capture, with charger voltage and instantaneous current values recorded directly from the battery service.

5.6 Network Configuration

Purpose: Captures the device's active network interfaces and live connections, useful for establishing what the device was communicating with, and over what kind of network, at the moment of acquisition.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell dumpsys connectivity
NetworkProviders for:
  1: EthernetNetworkFactory
  2: L2capNetworkProvider
  5: PhoneSwitcherNetworkRequestListener
  8: WifiNetworkFactory
  10: RestrictedWifiNetworkFactory
  9: UntrustedWifiNetworkFactory
  4: VpnNetworkProvider:0
  6: TelephonyNetworkFactory[0]
  7: TelephonyNetworkFactory[1]
  12: MultiInternetWifiNetworkFactory
  3: VcnNetworkProvider
  11: OemPaidWifiNetworkFactory

Active default network: 555

Current network preferences:
Default requests:

Current Networks:
NetworkAgentInfo(network[468] handle(2013450391565) ni(MOBILE[LTE] CONNECTED extra: ims) created-2026-07-24T06:56:51.128Z Score(Policies : EVER_EVALUATE
EDNEVER_VALIDATED&IS_VALIDATED ; KeepConnected : 0) created 1111527727 firstValidated 1111527895 lastValidated 1111527895 Ip({InterfaceName: rmnet_data3
```

Figure 13. adb shell dumpsys connectivity / wifi summary of the active network state.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell ifconfig
rmnet_data2 Link encap:UNSPEC
  inet6 addr: [REDACTED] Scope: Global
  UP RUNNING MTU:1500 Metric:1
  RX packets:185246 errors:0 dropped:0 overruns:0 frame:0
  TX packets:159306 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:1000
  RX bytes:134608866 TX bytes:42560604

dummy0 Link encap:UNSPEC
  inet6 addr: fe80::c69:f7ff:fec0:ad76/64 Scope: Link
  UP BROADCAST RUNNING NOARP MTU:1500 Metric:1
  RX packets:0 errors:0 dropped:0 overruns:0 frame:0
  TX packets:15782 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:1000
  RX bytes:0 TX bytes:3137687

rmnet_data0 Link encap:UNSPEC
  inet6 addr: fe80::c3d0:284e:fe5a:3dc3/64 Scope: Link
  UP RUNNING MTU:1500 Metric:1
  RX packets:4509 errors:0 dropped:0 overruns:0 frame:0
  TX packets:3054 errors:0 dropped:0 overruns:0 carrier:0
  collisions:0 txqueuelen:1000
  RX bytes:357974 TX bytes:292655
```

Figure 14. adb shell ifconfig, listing network interfaces and their addresses. The device's global-scope IPv6 address has been redacted (solid black bar) to protect the examiner's personal network identity.


```

C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell netstat
Active Internet connections (w/o servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp      0      1 10.88.250.121:47532    10.88.250.121:47532    FIN_WAIT1
tcp      0      1 192.168.1.8:40712     142.251.150.119:https  LAST_ACK
tcp      0 1752 192.168.0.100:48694   172.217.119.4:https    FIN_WAIT1
tcp      0      1 192.168.0.100:37132   119.153.112.112:https  FIN_WAIT1
tcp      0      1 192.168.1.10:46786    216.239.32.223:https   FIN_WAIT1
tcp      0 2070 192.168.0.100:45132   104.18.33.206:https    FIN_WAIT1
tcp      0      1 192.168.1.4:49180     172.217.117.4:https    FIN_WAIT1
tcp      0      1 192.168.1.10:39138    104.18.41.41:https     FIN_WAIT1
tcp      0      1 192.168.1.4:41506     142.251.157.1:https    FIN_WAIT1
tcp      0      1 192.168.0.100:40278   160.79.104.10:https    FIN_WAIT1
tcp      0      1 192.168.1.5:42100     172.217.118.4:https    FIN_WAIT1
tcp      0      1 192.168.0.100:41256   52.216.62.138:https    FIN_WAIT1
tcp      0      1 192.168.0.101:48102   172.64.146.215:https   FIN_WAIT1
tcp      0      1 192.168.1.5:48396     pnfjra-af-in-f5.1:https FIN_WAIT1
tcp      0      1 192.168.1.11:47344    142.251.152.119:https  LAST_ACK
tcp      0      1 192.168.1.6:48398     160.79.104.10:https    FIN_WAIT1
tcp      0      1 192.168.1.10:37690    pnfjra-ak-in-f3.1:https FIN_WAIT1
tcp      0      1 192.168.0.101:49814   172.64.146.215:https   FIN_WAIT1
tcp      0      1 192.168.1.5:41212     172.64.146.215:https   FIN_WAIT1
tcp      0 1734 192.168.0.100:45628   172.217.115.4:https    FIN_WAIT1
tcp      0      1 192.168.1.10:40406    172.217.114.4:https    FIN_WAIT1
tcp      0      1 192.168.1.10:40406    172.217.113.4:https    FIN_WAIT1
tcp      0      1 192.168.0.101:44982   119.156.245.16:https   FIN_WAIT1
tcp      0      1 192.168.1.10:44090    216.239.32.223:https   FIN_WAIT1
tcp      0      1 192.168.0.112:38178   104.18.41.41:https     FIN_WAIT1
tcp      1 273 192.168.1.9:47247     185.157.244.160:53955  CLOSING
tcp      0      1 192.168.1.5:38000     119.156.245.12:https   FIN_WAIT1
tcp      0      1 192.168.1.8:37264     142.251.156.119:https  LAST_ACK

```

Figure 15. adb shell netstat, part 1: active TCP connections by local/foreign address and state. A recurring public IPv4 address has been redacted throughout.

```

tcp      130    0 192.168.0.100:49908    172.217.116.4:https    CLOSE_WAIT
tcp      1      1 192.168.1.5:37910     119.153.112.112:https  CLOSING
tcp      0      1 192.168.1.10:42368    142.251.152.4:https    FIN_WAIT1
tcp      1 423 192.168.100.242:48672  pnfjra-al-in-f5.1:https CLOSING
tcp      0      1 192.168.1.10:37604    160.79.104.10:https    FIN_WAIT1
tcp      0      0 192.168.0.100:40686    172.64.146.215:https   ESTABLISHED
tcp      0      1 192.168.1.8:44014     216.239.38.223:https   FIN_WAIT1
tcp      0      1 192.168.1.8:38642     172.217.119.4:https    FIN_WAIT1
tcp      0 1531 10.88.250.121:37652    104.16.248.249:https   FIN_WAIT1
tcp      0      1 192.168.0.106:45786    172.64.146.215:https   FIN_WAIT1
tcp      130    0 192.168.0.100:41644    10.88.250.121:47532    CLOSE_WAIT
tcp      0      1 192.168.1.4:41512     142.251.157.1:https    FIN_WAIT1
tcp      0      1 192.168.1.5:41254     pnfjra-ag-in-f14.:https FIN_WAIT1
tcp      0      1 192.168.1.10:47194    172.217.115.4:https    FIN_WAIT1
tcp      0      1 192.168.1.11:40634    172.217.119.4:https    FIN_WAIT1
tcp      0      1 192.168.1.8:38142     pnfjra-aq-in-f14.:https FIN_WAIT1
tcp      0      0 192.168.0.100:44838    edge-dgw-mini-shv:https ESTABLISHED
tcp      0      1 192.168.0.100:47588    sin26s14-in-f9.1e:https FIN_WAIT1
tcp      0      1 192.168.1.6:48394     160.79.104.10:https    FIN_WAIT1
tcp      0      1 192.168.1.5:39378     119.153.111.17:https   FIN_WAIT1
tcp      0      1 192.168.1.8:40046     pnfjra-ao-in-f14.:https FIN_WAIT1
tcp      0      1 192.168.1.4:41510     142.251.157.1:https    FIN_WAIT1
tcp      0      1 192.168.0.100:39138    a23-44-5-26.deplo:https FIN_WAIT1
tcp      0      1 192.168.1.8:38656     216.239.34.223:https   FIN_WAIT1
tcp      0      1 192.168.1.101:40496    142.251.152.119:https  LAST_ACK
tcp      0      1 192.168.0.100:41702    142.251.153.119:https  LAST_ACK
tcp      0      1 192.168.0.100:49564    119.156.245.48:https   FIN_WAIT1

```

Figure 16. adb shell netstat, part 2, continuing the connection listing, with the same public IPv4 address redacted.

Finding: Two connections were captured in an ESTABLISHED state (to Cloudflare and Meta edge infrastructure); the remainder were in FIN_WAIT1, LAST_ACK, or CLOSE_WAIT, meaning they were already in the process of closing rather than actively open at the moment of capture.

TECHNICAL NOTE: TECHNICAL NOTE: READING NETSTAT CONNECTION STATES

ESTABLISHED means a connection was genuinely live and open at the exact moment netstat ran, both sides had an active session. FIN_WAIT1, LAST_ACK, and CLOSE_WAIT all describe a connection somewhere in TCP's multi-step closing handshake: the session had already been torn down or was actively being torn down, and would disappear from the list entirely within seconds.

For an examiner, this distinction matters: an ESTABLISHED entry is strong evidence of an active session at a specific timestamp, while a closing-state entry only proves a connection existed recently, not that it was still meaningfully in use. The two ESTABLISHED connections captured here are the most forensically significant entries in this network snapshot for exactly that reason; everything else is a trailing record of very recent, already-finished activity.

5.7 Directory and File Acquisition

Purpose: Tests direct file-level acquisition of accessible storage locations, and, separately, of a protected system path, to demonstrate both what ADB can retrieve without root and where it is blocked.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb pull /sdcard/DCIM ./Android_Images
[ 67%] /sdcard/DCIM/Camera/IMG20241229151342.jpg: 50%
```

Figure 17. adb pull of /sdcard/DCIM to a local Android_Images folder, in progress.

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>dir Android_Images
Volume in drive C is Windows
Volume Serial Number is 8457-0DC6

Directory of C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools\Android_Images

26/07/2026  12:21  am  <DIR>      .
26/07/2026  12:21  am  <DIR>      ..
26/07/2026  12:21  am  <DIR>      .convert_tmp_files
26/07/2026  12:21  am             89 .shareit_beyla_ids.cfg
26/07/2026  12:21  am             85 .sys_install_s_time.cfg
26/07/2026  12:33  am  <DIR>      Camera
26/07/2026  12:21  am  <DIR>      CamScanner
26/07/2026  12:21  am  <DIR>      Duolingo
26/07/2026  12:21  am  <DIR>      Facebook
26/07/2026  12:21  am  <DIR>      JazzCash
26/07/2026  12:21  am  <DIR>      LightroomCamera
26/07/2026  12:21  am  <DIR>      MCB
26/07/2026  12:21  am  <DIR>      MyAlbums
26/07/2026  12:21  am  <DIR>      Restored
26/07/2026  12:21  am  <DIR>      Screenshots
26/07/2026  12:21  am  <DIR>      WhatsApp
                2 File(s)                174 bytes
                14 Dir(s)  142,481,981,440 bytes free
```

Figure 18. Local directory listing of the pulled DCIM contents: camera photos alongside app-specific media folders (JazzCash, MCB, Facebook, WhatsApp, Duolingo, CamScanner).

```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb pull /sdcard/Download ./Downloads
[ 2%] /sdcard/Download/the-book-of-wisdom-a-collection-of-timeless-wisdom-from-around-the-world.pdf: 28%
```

Figure 19. adb pull of /sdcard/Download to a local Downloads folder, in progress.

```

C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>dir Downloads
Volume in drive C is Windows
Volume Serial Number is 8457-0DC6

Directory of C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools\Downloads

26/07/2026  12:34 am  <DIR>          .
26/07/2026  12:34 am  <DIR>          ..
26/07/2026  12:34 am             444,118 #04_ General Knowledge-4.pdf
26/07/2026  12:34 am             469,231 #3 General Knowledge-3.pdf
26/07/2026  12:34 am             423,293 #5 General Knowledge Test -5.pdf
26/07/2026  12:34 am  <DIR>          .com_coloros_smartsidebar
26/07/2026  12:34 am              32 .exmu-cfg1.data
26/07/2026  12:34 am             227,258 10639534-8683-4f85-93b7-306e879c2694.jpeg
26/07/2026  12:34 am             136,596 1765280061897.pdf
26/07/2026  12:34 am             8,324,824 1781861646713.pdf
26/07/2026  12:34 am  <DIR>          1DMP
26/07/2026  12:34 am             971,329 240600272.pdf
26/07/2026  12:34 am             841,706 250503112 (1).pdf
26/07/2026  12:34 am              0 250503112.pdf
26/07/2026  12:34 am              0 251000585 (1).pdf
26/07/2026  12:34 am              0 251000585.pdf
26/07/2026  12:34 am             270,137 740905_1.jpg
26/07/2026  12:34 am             639,112 Abubakr Latif-Rollno-Slip.pdf
26/07/2026  12:34 am             439,332 ADVANCE_ELECTRICAL_POWER_SYSTEM_6_Months.zip
26/07/2026  12:34 am             642,352 Awais Ahmed-Rollno-Slip.pdf
26/07/2026  12:34 am              5,208 AwaisAhmed_AdminResume.pdf

```

Figure 20. Local directory listing of the pulled Downloads contents: academic documents, a resume, and other personal files.

Finding: Both /sdcard/DCIM and /sdcard/Download were fully accessible without root, and were pulled successfully; the resulting local folders reflect ordinary personal device use, camera photos, e-wallet and banking app storage folders, and personal/academic documents, with nothing indicating anomalous or hidden activity.

A further attempt was made to pull a protected system file directly:

```

C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb pull /data/system/packages.list ./Package_list
adb: error: failed to stat remote object '/data/system/packages.list': Permission denied

```

Figure 21. adb pull of /data/system/packages.list, failing with Permission denied.

Finding: Access to /data/system/packages.list was denied. Paths under /data are protected by Android's application sandboxing and are not accessible over a standard (non-root) ADB shell; retrieving them would require a rooted device or a forensic-mode system image, which is why this step returned no result rather than partial data.

5.8 Device Timeline and Activity Data

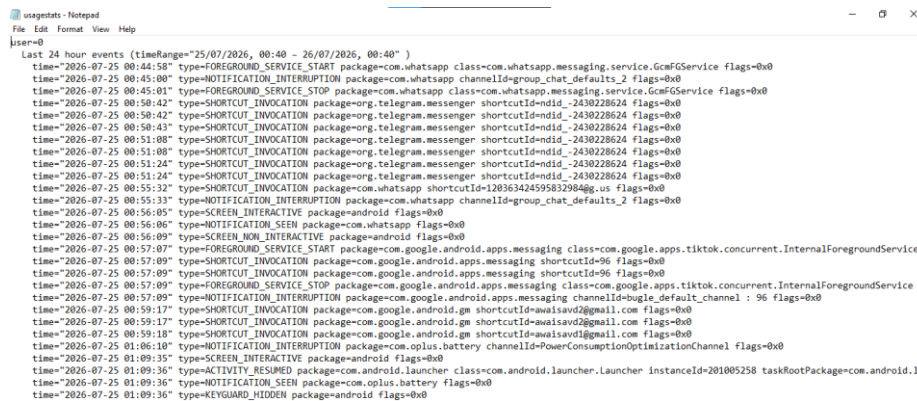
Purpose: Reconstructs recent usage patterns and system-level state from Android's own usage-tracking and settings services, without needing access to the restricted content providers covered in Section 6.

```

C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell dumpsys usagstats > usagstats.txt

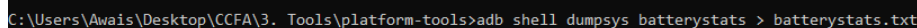
```

Figure 22. Redirecting adb shell dumpsys usagstats to usagstats.txt.



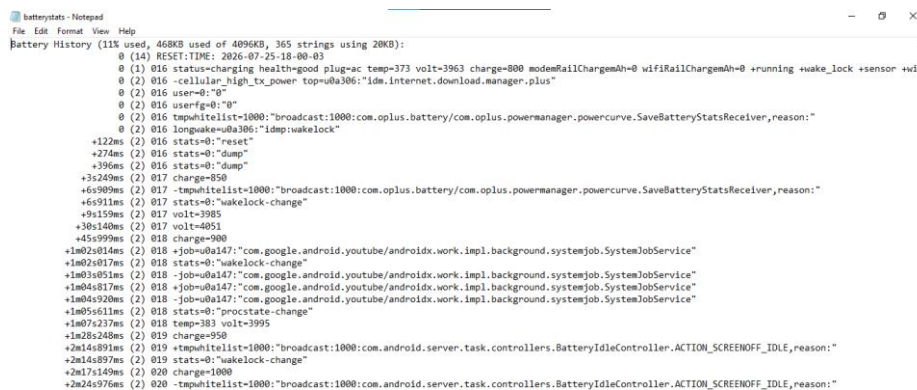
```
usagstats - Notepad
File Edit Format View Help
Last 24 hour events (TimeRange="25/07/2026, 00:40 - 26/07/2026, 00:40")
jser=0
time="2026-07-25 00:44:58" type=BACKGROUND_SERVICE_START package=com.whatsapp class=com.whatsapp.messaging.service.GcmFService flags=0x0
time="2026-07-25 00:45:00" type=NOTIFICATION_INTERRUPT package=com.whatsapp channelId=group_chat_defaults_2 flags=0x0
time="2026-07-25 00:45:01" type=BACKGROUND_SERVICE_STOP package=com.whatsapp class=com.whatsapp.messaging.service.GcmFService flags=0x0
time="2026-07-25 00:50:42" type=SHORTCUT_INVOCATION package=org.telegram.messenger shortcutId=ndid_-2430228624 flags=0x0
time="2026-07-25 00:50:42" type=SHORTCUT_INVOCATION package=org.telegram.messenger shortcutId=ndid_-2430228624 flags=0x0
time="2026-07-25 00:51:08" type=SHORTCUT_INVOCATION package=org.telegram.messenger shortcutId=ndid_-2430228624 flags=0x0
time="2026-07-25 00:51:08" type=SHORTCUT_INVOCATION package=org.telegram.messenger shortcutId=ndid_-2430228624 flags=0x0
time="2026-07-25 00:51:24" type=SHORTCUT_INVOCATION package=org.telegram.messenger shortcutId=ndid_-2430228624 flags=0x0
time="2026-07-25 00:51:24" type=SHORTCUT_INVOCATION package=org.telegram.messenger shortcutId=ndid_-2430228624 flags=0x0
time="2026-07-25 00:55:32" type=SHORTCUT_INVOCATION package=com.whatsapp shortcutId=128363424595832984@us flags=0x0
time="2026-07-25 00:55:33" type=NOTIFICATION_INTERRUPT package=com.whatsapp channelId=group_chat_defaults_2 flags=0x0
time="2026-07-25 00:56:05" type=SCREEN_INTERACTIVE package=android flags=0x0
time="2026-07-25 00:56:06" type=NOTIFICATION_SEEN package=com.whatsapp flags=0x0
time="2026-07-25 00:56:09" type=SCREEN_NON_INTERACTIVE package=android flags=0x0
time="2026-07-25 00:57:07" type=BACKGROUND_SERVICE_START package=com.google.android.apps.messaging class=com.google.apps.tiktok.concurrent.InternalForegroundService
time="2026-07-25 00:57:09" type=SHORTCUT_INVOCATION package=com.google.android.apps.messaging shortcutId=96 flags=0x0
time="2026-07-25 00:57:09" type=SHORTCUT_INVOCATION package=com.google.android.apps.messaging shortcutId=96 flags=0x0
time="2026-07-25 00:57:09" type=BACKGROUND_SERVICE_STOP package=com.google.android.apps.messaging class=com.google.apps.tiktok.concurrent.InternalForegroundService
time="2026-07-25 00:57:09" type=NOTIFICATION_INTERRUPT package=com.google.android.apps.messaging channelId=bugle_default_channel : 96 flags=0x0
time="2026-07-25 00:59:17" type=SHORTCUT_INVOCATION package=com.google.android.gm shortcutId=awaisavd2@gmail.com flags=0x0
time="2026-07-25 00:59:17" type=SHORTCUT_INVOCATION package=com.google.android.gm shortcutId=awaisavd2@gmail.com flags=0x0
time="2026-07-25 00:59:18" type=SHORTCUT_INVOCATION package=com.google.android.gm shortcutId=awaisavd1@gmail.com flags=0x0
time="2026-07-25 00:59:18" type=NOTIFICATION_INTERRUPT package=com.oplus.battery channelId=PowerConsumptionOptimizationChannel flags=0x0
time="2026-07-25 01:09:35" type=SCREEN_INTERACTIVE package=android flags=0x0
time="2026-07-25 01:09:36" type=ACTIVITY_RESUMED package=com.android.launcher class=com.android.launcher.Launcher instanceId=201805258 taskRootPackage=com.android.l
time="2026-07-25 01:09:36" type=NOTIFICATION_SEEN package=com.oplus.battery flags=0x0
time="2026-07-25 01:09:36" type=KEYGUARD_HIDDEN package=android flags=0x0
```

Figure 23. usagstats.txt excerpt, showing recent app and shortcut activity, including Gmail account shortcut identifiers.



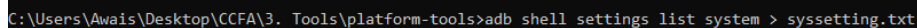
```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell dumsys batterystats > batterystats.txt
```

Figure 24. Redirecting adb shell dumsys batterystats to batterystats.txt.



```
batterystats - Notepad
File Edit Format View Help
Battery History (11K used, 468KB used of 4096KB, 365 strings using 20KB):
0 (14) RESET:TIME: 2026-07-25-18:00-03
0 (1) 016 status=charging health=good plug=ac temp=373 volt=3963 charge=800 modeMailChargedAh=0 wifiRailChargedAh=0 +running +wake_lock +sensor +wi
0 (2) 016 -cellular_high_tx_power top=0a306:ldm.internet.download.manager.plus"
0 (2) 016 user=0:"0"
0 (2) 016 userf=0:"0"
0 (2) 016 tmpwhitelist=1000:"broadcast:1000:com.oplus.battery/com.oplus.powermanager.powercurve.SaveBatteryStatsReceiver,reason:"
0 (2) 016 longwake=0a306:"ldm:wakeLock"
+132ms (2) 016 stats=0:"reset"
+274ms (2) 016 stats=0:"dump"
+396ms (2) 016 stats=0:"dump"
+3c249ms (2) 017 charge=850
+6s909ms (2) 017 -tmpwhitelist=1000:"broadcast:1000:com.oplus.battery/com.oplus.powermanager.powercurve.SaveBatteryStatsReceiver,reason:"
+6s91ms (2) 017 stats=0:"wakeLock-change"
+9s159ms (2) 017 volt=3985
+30s140ms (2) 017 volt=4051
+45s999ms (2) 018 charge=900
+1m02s014ms (2) 018 -job=0a147:"com.google.android.youtube/androidx.work.impl.background.systemjob.SystemJobService"
+1m02s017ms (2) 018 stats=0:"wakeLock-change"
+1m03s051ms (2) 018 -job=0a147:"com.google.android.youtube/androidx.work.impl.background.systemjob.SystemJobService"
+1m04s017ms (2) 018 -job=0a147:"com.google.android.youtube/androidx.work.impl.background.systemjob.SystemJobService"
+1m04s030ms (2) 018 -job=0a147:"com.google.android.youtube/androidx.work.impl.background.systemjob.SystemJobService"
+1m05s61ms (2) 018 stats=0:"procstate-change"
+1m07s237ms (2) 018 temp=383 volt=3995
+1m28s240ms (2) 019 charge=950
+2m14s091ms (2) 019 -tmpwhitelist=1000:"broadcast:1000:com.android.server.task.controllers.BatteryIdleController.ACTION_SCREENOFF_IDLE,reason:"
+2m14s097ms (2) 019 stats=0:"wakeLock-change"
+2m17s149ms (2) 020 charge=1000
+2m24s976ms (2) 020 -tmpwhitelist=1000:"broadcast:1000:com.android.server.task.controllers.BatteryIdleController.ACTION_SCREENOFF_IDLE,reason:"
```

Figure 25. batterystats.txt excerpt: a timestamped battery history log of charge level, screen state, and background job activity.



```
C:\Users\Awais\Desktop\CCFA\3. Tools\platform-tools>adb shell settings list system > syssetting.txt
```

Figure 26. Redirecting adb shell settings list system to syssetting.txt.



```
syssetting - Notepad
File Edit Format View Help
3GMode_on=0
ALARM_STATE=0
COM_PLATFORM_ACCOUNT_USERCENTER_ACCOUNT_BASIC_DATA={"classifyAge":"ADULT","country":"PK","se":"N5Tm3u31AwrZTum8FIkhz2BkmJ7d8srk8WmtatkiSkiah2VeSujMj7hDe7hQgd0z0k8ukCtkC
COM_PLATFORM_ACCOUNT_USERCENTER_ACCOUNT_COUNTRY=PK
DarkMode_BackgroundMax_System=0
DarkMode_DialogMax_System=27
DarkMode_ForegroundMini_System=100
DarkMode_style_key=0
DialogData=0
DialogDataMultiKey=0
GPSSmode_fisrt=0
KEYGUARD_VERSION=2
OPULUS_BATTERY_DATE_CHANGE_TIME=1785080000553
OPULUS_SIM_VSIM_ID=-1
PASSWORD_LENGTH=6
QTGAME_PKG=
SETTINGS_ACCOUNT_SOK_CONFIG_KEY={"defaultControl":{"errorFreq":60000,"errorCount":10},"authApiControl":{"successCount":2,"successFreq":120000,"errorFreq":60000,"errorC
SLA_DIALOG_COUNT=3
SpeedControlPara=ver=num;1;enable=false;thermal=40;limit_level=80,40,10,1;limit_level_up=10,4,2,1;prohibitbittimer=0
USERCENTER_ACCOUNT_LOGIN=1
USERCENTER_ACCOUNT_TOKEN_HASH=2098761282
USERCENTER_ACCOUNT_USERINFO_HASH=1761850222
accelerometer_rotation=0
accelerometer_rotation_secondary=0
access_color_setting=-1
adfr_factory_mode=0
alarm_alert-content://media/internal/audio/media/317?title=ringtone_035&canonical=1
alarm_alert_set=1
apply_ramping_ringer=0
auto_answer=1
```

Figure 27. syssetting.txt excerpt, showing an OPPO account entry with country code and an account session token.

Finding: Usage-stats data surfaced the device's two configured Gmail accounts (awaisavd2@gmail.com, awaisavd1@gmail.com) via their shortcut identifiers. Battery history provided a fine-grained, second-by-second timeline of charge level and background activity. System settings confirmed an active OPPO account tied to Pakistan (PK) as the registered country, along with its session token.

6. Where Android 13 Blocks Further Access

On Android 11 and higher, ADB shell access to contacts, call logs, and SMS is restricted unless the device is rooted or a special forensic build is used, exactly the same protection boundary already encountered directly in Section 5.7's failed /data pull. Because the examiner's device is an unmodified, non-rooted Android 13 phone, the commands below could not be executed against it in this exercise. They are documented here as reference material: this is what a rooted device or a forensic-mode image would additionally make available, not a record of data actually recovered in this exercise.

TECHNICAL NOTE: TECHNICAL NOTE: DOCUMENTED BUT NOT EXECUTED

Every command in this section targets a Content Provider (Android's structured inter-app data-access layer), rather than a plain file, which is precisely the layer Android 11+ locks down for third-party and shell access. A rooted device, or an official forensic acquisition build, bypasses this restriction because it runs the ADB shell with elevated (system/root) privileges instead of the standard, sandboxed shell user.

List All Applications with Registered Accounts

```
adb shell dumpsys account | grep -i com.*$ -o | cut -d' ' -f1 | cut -d} -f1 | grep -v com$
```

Lists all app package names that have registered accounts on the device.

List Email Addresses Registered on the Device

```
adb shell dumpsys | grep -E -o "\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,6}\b"
```

Extracts every detected email address from the Account Manager service.

Count Number of Device Reboots

```
adb shell settings list global | grep "boot_count=" | cut -d= -f2 | head -n 1 | xargs echo "Booted:" | sed 's/$/ times/g'
```

Retrieves the device boot counter from global system settings.

List Every Contact and Phone Number

```
adb shell content query --uri content://contacts/phones/ --projection display_name:number | cut -f 3- -d " "
```

Shows all stored contact names and phone numbers.

Extract All Contact Info

```
adb shell content query --uri content://contacts/phones/
```

Lists raw contact provider data for quick inspection.

Dump Call Log

```
adb shell content query --uri content://call_log/calls
```

Retrieves call history entries including number, type, and timestamp.

Dump SMS Messages

```
adb shell content query --uri content://sms/
```

Exports SMS database contents such as address, date, and body, redirected to sms.txt on a device where this provider is accessible.

7. Findings Summary and Conclusion

TECHNICAL NOTE: OVERALL ASSESSMENT

Device identification (model, Android version, serial number) was recovered cleanly and matches the known physical device throughout.

Installed applications, real-time logs, and battery/power state were all fully accessible without root, and reflect ordinary, expected personal device use with no anomalous findings.

Network configuration captured two genuinely live (ESTABLISHED) connections at the moment of acquisition, distinct from several already-closing sessions, a distinction that matters for any timeline built from this data.

Accessible storage (DCIM, Download) was pulled successfully; the protected /data path correctly returned Permission denied, demonstrating Android's sandboxing working as intended against a non-rooted shell.

Usage statistics, battery history, and system settings together reconstructed a detailed activity timeline without needing access to the restricted contacts/call-log/SMS providers covered in Section 6.

Because this device is an unmodified, non-rooted Android 13 phone, this exercise reached the practical ceiling of what standard ADB access can recover: everything documented in Section 5 was retrieved directly, while contacts, call logs, and SMS, along with protected system files, remained out of reach and are only documented as reference commands in Section 6. On a rooted device or with a proper forensic-mode image, the same command set would extend directly into those restricted categories with no change in method, only in the privilege level the shell is running under.

As a self-examination exercise, this report also demonstrates the flip side of mobile forensics: exactly how much of a person's daily activity, accounts, app usage, network behavior, and file history, a standard, non-rooted ADB session can reconstruct without ever touching the device's most protected data stores.